GREEDY ALGORITHMS

1-G-Coin Problem

```
#include <stdio.h>
int main() {
    int v;
    int count = 0;
    scanf("%d", &v);
    int coins[] = {1000, 500, 100, 50, 20, 10, 5, 2, 1};
    for (int i = 0; i < 9; i++) {
        count = count + v / coins[i];
        v = v % coins[i];
    }
    printf("%d", count);
    return 0;
}
```

2-G-Cookies Problem

```
#include <stdio.h>
int main() {
    int n, m;
    scanf("%d", &n);
    int g[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &g[i]);
    }
    scanf("%d", &m);
```

```
int s[m];
for (int i = 0; i < m; i++) {
    scanf("%d", &s[i]);
}
for (int i = 0; i < n - 1; i++) {
    for (int j = i + 1; j < n; j++) {
        if (g[i] > g[j]) {
            int temp = g[i];
            g[i] = g[j];
            g[j] = temp;
        }
    }
}
for (int i = 0; i < m - 1; i++) {
    for (int j = i + 1; j < m; j++) {
        if (s[i] > s[j]) {
            int temp = s[i];
            s[i] = s[j];
            s[j] = temp;
        }
    }
}
int i = 0;
int j = 0;
int count = 0;
while (i < n && j < m) {
    if (s[j] >= g[i]) {
        count++;
        i++;
```

```
            j++;
        } else {
            j++;
        }
    }
    printf("%d", count);
    return 0;
}
```

3-G-Burger Problem

```
#include <stdio.h>
#include <math.h>
int main()
{
    int n;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    for (int i = 0; i < n - 1; i++)
    {
        for (int j = 0; j < n - i - 1; j++)
        {
            if (arr[j] < arr[j + 1])
            {
```

```
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    int s = 0;
    for (int i = 0; i < n; i++)
    {
        s += (int)(pow(n, i)) * arr[i];
    }
    printf("%d", s);
    return 0;
}
```

4-G-Array Sum max problem

```
#include <stdio.h>
#include <stdlib.h>
int compare(const void *a, const void *b) {
    return (*(int *)a - *(int *)b);
}
int main() {
    int n;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
```

```
    qsort(arr, n, sizeof(int), compare);
    long long sum = 0;
    for (int i = 0; i < n; i++) {
        sum += (long long)arr[i] * i;
    }
    printf("%lld", sum);
    return 0;
}
```

5-G-Product of Array elements-Minimum

```
#include <stdio.h>
int main() {
    int n, i, j, temp;
    long long sum = 0;
    scanf("%d", &n);
    int A[n], B[n];
    for (i = 0; i < n; i++)
        scanf("%d", &A[i]);
    for (i = 0; i < n; i++)
        scanf("%d", &B[i]);
    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (A[i] > A[j]) {
                temp = A[i];
                A[i] = A[j];
                A[j] = temp;
            }
        }
```

```
    }
    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (B[i] < B[j]) {
                temp = B[i];
                B[i] = B[j];
                B[j] = temp;
            }
        }
    }
    for (i = 0; i < n; i++)
        sum += (long long)A[i] * B[i];
    printf("%lld", sum);
    return 0;
}
```